Kangnan Ye

San Diego, California, United States, 92122
(858)-828-6299 | kaye@ucsd.edu

EDUCATION

University of California - San Diego Sep 2023 - Jun 2027 (Expected)
Major: Engineering Physics; Physics with specialization in Material Science, Minor: Math

TECHNICAL SKILLS

- **Software**: Matlab, Python, Cadence, Vivado, Pspice, Verilog, ImageJ
- **Hardware**: CNT coating, Circuit Building, Arduino
- **Coursework:** Newtonian Mechanics, Fluid Mechanics, Quantum Mechanics, Electromagnetism, ODEs, Analysis, Circuit (Amplifier Circuits, CMOS, Digital Circuit) design, Circuit analysis, Semiconductor Physics, Solid State Physics

RESEARCH EXPERIENCES

Lower-Back Pressure Asymmetry – **ARMOR Lab, UC San Diego** *June 2025 – Present*

- Collected and analyzed synchronized EMG and motion-capture (MoCap) data during postural and squat-based movements.
- Developed a signal-processing and analysis pipeline to time-align EMG and MoCap-derived strain metrics, enabling correlation analysis and identification of timing discrepancies.
- Investigating differences between electrical muscle activation (EMG) and mechanical strain measures to inform future design of integrated EMG–CNT wearable sensors.
- Designed and prototyped a hybrid EMG–CNT sensing system for concurrent measurement of muscle activation and surface strain.
- Engineered sensor layering and interfacing considerations (electrode contact, strain-sensitive CNT layer, insulation) to enable multimodal signal acquisition in a wearable format.

Sentiment Analysis of Patient-Doctor Communication – **Project, UC San Diego** *June 2025 – Present*

- Trained BERT and DistilBERT models to detect emotional tone in anonymized clinical messages; evaluated fine-tuning versus frozen-embedding performance.
- Integrated TF-IDF + logistic regression baselines with K-means clustering for unsupervised comparison of semantic similarity patterns.
- Compared cluster composition and classifier outputs to assess consistency between semantic similarity and emotional tone.
- Automated the pipeline in Python (PyTorch, scikit-learn, pandas) and exported labeled datasets for research collaborators.

Digital Circuit Design and CMOS Layout Optimization – **ECE 165 Project, UC San Diego** Mar 2025 – June 2025

- Implemented a Ladner–Fischer adder in Verilog, achieving logarithmic carry propagation and reduced delay.
- Designed a transmission-gate-based CMOS layout in Cadence Virtuoso with customized transistor sizing to balance rise/fall times.
- Verified post-layout performance improvement (~20 % faster propagation) through timing and parasitic-extraction analysis.

Recursive Karatsuba Multiplication in LEGv8 Assembly – **ECE 30 Project, UC San Diego** Mar 2025 – June 2025

- Conducted runtime benchmarking and correctness verification using controlled test cases, demonstrating a ~30 % execution-time reduction compared to naïve long multiplication.
- Designed low-level subroutines for partial-product computation, modular addition, and carry propagation, ensuring numerical stability across recursive calls.
- Debugged and verified arithmetic correctness through iterative testing and cycle-count benchmarking in the LEGv8 simulator.